

# ■ ESP32 Smart Grid Monitoring & Fault Analysis System

Comprehensive University Master Textbook & Reverse Engineering Manual — Phase 1

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## Phase 1: Project Overview & Strategic Foundations

### 1. What Problem Does This Project Solve?

Modern electrical distribution networks face severe operational challenges due to traditional 'dumb' infrastructure. When an outage, voltage sag, or short circuit occurs, utility companies historically rely on manual customer phone complaints to discover that a power line has failed. This causes massive delays in fault location, equipment destruction from unmitigated overload spikes, and severe financial losses. This project solves this by placing IoT sensor nodes (ESP32) on power lines that continuously stream 3-phase voltage, current, active/reactive power, power factor, and frequency data every 1 second over MQTT to a central Python server. The server instantly calculates AC power flow physics (Pandapower), evaluates short-circuit fault currents and relay trip times (OpenDSS), and executes AI models (IsolationForest & RandomForest) to catch anomalies and predict 24-hour demand.

### 2. Why is This Project Important?

Power distribution grids are undergoing the global '3Ds Energy Transition': Decarbonization (shifting to solar/wind), Decentralization (rooftop solar feeding power back into local feeders), and Digitalization. Without digital monitoring, reverse power flows from rooftop solar panels cause distribution line overvoltages and transformer overheating. This project is critical because it provides an open-source, sub-second telemetry and AI intelligence platform at 1% of the cost of legacy industrial SCADA systems.

### 3. Real World Applications

- **Smart Cities:** Neighborhood 3-phase voltage balance and automated transformer health surveillance.
- **Solar & Wind Microgrids:** Tracking reverse energy feed-in and managing solar intermittency.
- **Industrial Plants:** Monitoring heavy motor power factor ( $\cos \phi \geq 0.90$ ) to prevent utility penalty charges.
- **EV Fast-Charging Hubs:** Dynamic load management to prevent substation transformer overloads.

### 4. Target Users

- **Electrical Distribution Utilities (Discoms):** Control room operators monitoring substations.
- **Microgrid & Renewable Energy Engineers:** Managing localized solar PV farms.
- **Factory Energy Managers:** Preventing industrial motor burnout from low voltage sags.
- **Smart City Municipal Engineers:** Municipal power efficiency and streetlight telemetry.

### 5 & 6. Existing Solutions & Limitations of Traditional Systems

Existing industrial SCADA solutions (Siemens Spectrum Power, Schneider EcoStruxure, ABB Ability, GE GridIQ) cost \$500,000 to \$5,000,000+ per substation. Traditional systems suffer from:

1. **High Cost:** Multi-million dollar hardware & software licensing.
2. **Slow Refresh Rates:** Legacy utility meters sample data only once every 15 to 60 minutes.
3. **Proprietary Lock-in:** Siemens software works only with Siemens hardware controllers.
4. **Siloed Architectures:** Physics engines and AI analytics are sold as separate, disconnected enterprise

applications.

## 7. Why This Approach is Better

Your project is superior because it uses standard open-source Python, lightweight MQTT Wi-Fi messaging (1-second latency), low-cost ESP32 microcontrollers (\$15 per node), and natively unifies non-linear AC Newton-Raphson power flow physics with Machine Learning AI inside a single high-density cyber web dashboard accessible on any mobile browser.

## 8 & 9. Business Value & Technical Value

**Business Value:** Saves utility companies millions by preventing transformer explosions, reducing outage repair times from hours to seconds, and eliminating reactive power penalty fees.

**Technical Value:** Combines Kirchhoff's electrical laws (Pandapower), OpenDSS sequence matrix fault math, and Scikit-Learn unsupervised IsolationForest anomaly scoring into a unified multi-threaded architecture.

## 10. Future Scope

1. **Hardware Relay Tripping:** Program ESP32 outputs to trigger physical circuit breaker relays on short circuits.
2. **LoRaWAN / 4G Cellular:** Deploy SIM800L LTE modems for remote rural power lines up to 15 km.
3. **LSTM Deep Learning:** Upgrade load forecasting to Long Short-Term Memory neural networks.
4. **Blockchain P2P Energy Trading:** Smart contracts allowing solar households to sell excess power to neighbors.